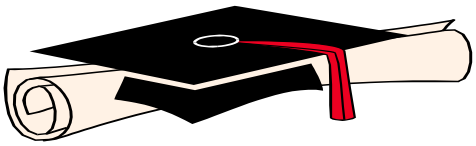
AP

Programming in C#

Semester: 1	**The Exam Code:** 03
Duration: 90 minutes	**Total Marks:** 20



Practical Examination Paper

Do not write on this question paper and return it to the Invigilator after the examination.

Instructions

- Write clean, well-structured code with meaningful comments.
- Validate user input and handle exceptions where necessary.
- Test your code to ensure it works correctly for various scenarios.

Question 1: Create an Interface (2 Points)

Create an interface `IEventTicket` with the following methods:

- `float CalculateTotalCost();` – Calculates the total cost of the tickets.
- `void DisplayTicketDetails();` – Displays all ticket information.

Question 2: Create the EventTicket Class (6 Points)

Task:

Create a class `EventTicket` that implements the `IEventTicket` interface.

Fields and Properties:

Field Name	Data Type	Description
ticketId	int	Unique identifier for the ticket
eventName	string	Name of the event (3-100 characters)
ticketPrice	float	Price per ticket (> 0)
quantity	int	Number of tickets purchased (> 0)
eventDate	DateTime	Date of the event (must be in the future)

Methods:

- `CalculateTotalCost()` (1.5 Points):
  `TotalCost = ticketPrice * quantity`

- DisplayTicketDetails() (1.5 Points):
  Example output:
  Concert "Jazz Night" on 2024-08-01: 3 tickets at $120 each, total = $360

Validation Rules (3 Points):

- eventName must be 3–100 characters.
- ticketPrice must be > 0.
- quantity must be > 0.
- eventDate must be a future date.

Question 3: Create the VIPEventTicket Class (4 Points)

Task:
Create a class VIPEventTicket that inherits from EventTicket.

Additional Field:

Field Name	Data Type	Description
vipChargeRate	float	Extra fee rate for VIP tickets (default: 20%)

Override Methods:

- CalculateTotalCost() (2 Points):
  TotalCost = (ticketPrice * quantity) + (ticketPrice * quantity * vipChargeRate / 100)
- DisplayTicketDetails() (2 Points):
  Example output:
  VIP Concert "Pop Live" on 2024-09-12: 2 tickets at $150 each, total = $360

Question 4: Implement Ticket Management Class (5 Points)

Task:
Create a class TicketManager to perform CRUD operations on a list of tickets.

Features:

- **Add Ticket** (1.5 Points):
  Add new EventTicket or VIPEventTicket to the list. Validate all input.

- **View Tickets** (1.5 Points):
  Display all tickets with details using DisplayTicketDetails().

- **Update Ticket** (1 Point):
  Allow update of ticketPrice and quantity using ticketId.

- **Delete Ticket** (1 Point):
  Remove a ticket using its ID. Show error message if not found.

Question 5: Main Test Class (3 Points)

Task:
Create a Program class with a menu-based system in the Main() method.

Menu Example:

```
=== EVENT TICKET MENU ===
1. Add Event Ticket / VIP Ticket
2. View All Tickets
3. Update Ticket
4. Delete Ticket
5. Calculate Total Cost
6. Exit
Your choice:
```

Additional:

- Prompt for ticket ID when using calculate cost option (1 Point).
- Catch and handle exceptions such as:
  - Invalid ticket ID
  - Negative price or quantity
  - Past event date (1 Point)

- Menu-driven structure with looping (1 Point)

 Submission Instructions:

- Submit all C# source code files (.cs) in a zipped folder named:
  CSharp_EventTicket_<YourName>.zip
- Make sure the code compiles and runs without errors.
- Include comments explaining your logic in each class/method.